

Econometrics of Financial Markets

Alessandro Bonfiglio 0001172511

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Question 1

The preliminary analysis shows that the returns do not exhibit autocorrelation in the mean (Figure 1), but show dependence in their conditional variance (Figure 2), which is the phenomenon of volatility clustering. The Ljung–Box test confirms this pattern, failing to detect autocorrelation in returns while rejecting the null hypothesis for volatility. The Jarque–Bera test, in turn, rejects normality in both cases, suggesting non-Gaussian distributions. The ADF test confirms the stationarity of the series. Descriptive statistics reveal mild leptokurtosis in the returns (kurtosis = 3.86) and negligible skewness (≈ 0), indicating heavier tails than the normal distribution but no dominant direction. Volatility, on the other hand, exhibits very high skewness and kurtosis values (10.9 and 213), reflecting an asymmetric distribution with rare yet intense spikes of turbulence. This behavior, not fully captured by the standard GARCH specification, motivates the adoption of the T-GARCH model with a Student's t distribution, which is better suited to model fat tails and the non-normality typical of financial data. Finally, the ARCH–LM test rejects the null hypothesis of homoskedasticity, providing additional evidence of conditional heteroskedasticity and confirming the suitability of ARCH-GARCH-type models. To model this behavior, the analysis begins with the ARCH model, which, while capturing the heteroskedastic nature of the series, depends solely on the most recent shock. This results in a highly erratic volatility pattern, as the model tends to underestimate volatility persistence (Figure 3). By introducing an autoregressive term and moving to the GARCH specification, shocks are allowed to decay gradually over time. Consequently, the volatility becomes smoother, displaying more persistent and prolonged periods of high and low variability (Figure 4). The Q–Q plot of the standardized residuals against the normal distribution shows that normality does not hold. This is typical in financial return series, where empirical distributions often display fat tails and some asymmetry, as negative shocks tend to increase volatility more than positive ones (Figure 5). To account for this, the model is extended by assuming that the errors follow a Student's t distribution. This change in the error distribution leads to an improvement in the log-likelihood, meaning that the t -distributed model explains the observed data more effectively with the same number of parameters, thanks to a more realistic distributional assumption. Finally, a residual diagnostic test confirms that the standardized residuals closely follow a theoretical t -distribution with approximately seven degrees of freedom, supporting the validity of the model specification (Figure 6). Additional support comes from the plot of standardized residuals and their conditional standard deviation bands. The bands widen during periods of high volatility, such as around 2008 and 2020, and narrow during calmer phases. Most importantly, the residuals remain largely within the bands, with only a few exceptional spikes, consistent with a distribution that allows for heavier tails than the normal. The residuals also follow the same pattern as the volatility bands over time, indicating that the model successfully captures the volatility clustering in the data (Figure 7). Another parameter worth examining is the sum of the coefficients, $\alpha + \beta$. In both models, this value is approximately 0.92, indicating that the improvement achieved by the T-GARCH model primarily concerns the distribution of shocks. The persistence of volatility remains almost unchanged, but the T-GARCH captures heavier and more leptokurtic tails, leading to an improvement in the log-likelihood (from 813.09 to 954.59). An EWMA model was also introduced as a benchmark: the estimated volatility follows the dynamics of the ARCH and GARCH models, though in a smoother way (Figure 8). The log-likelihood of this model, however, cannot be directly compared to the others, since, unlike parametric models, the smoothing parameter λ is fixed a priori (around 0.94, as in the RiskMetrics specification). When considering the half-life, a clear difference emerges: it increases from roughly eight days in the GARCH models (with the normal-distribution version slightly shorter than the t -distribution one, consistent with the presence of heavier tails) to about eleven days in the EWMA. This reflects the model's operational logic: the exponential weighting filter favors stability over reactivity, updating volatility gradually yet continuously. Regarding the diagnostic tests, both models (ARCH and GARCH) fail to reject the null hypothesis of the Ljung–Box test on standardized residuals, indicating the absence of linear autocorrelation (Figure 9 and Figure 10). The key improvement, however, lies in the GARCH model, where the p -value for the squared residuals is ≈ 0.59 , indicating that the residuals variance is no longer autocorrelated (Figure 11). This is a highly relevant result, given that the ARCH model previously rejected the null hypothesis, failing to capture the full dynamics of volatility (Figure 12). As discussed, the ARCH model adequately represents recent shocks but performs poorly in modeling volatility persistence. For completeness, the Jarque-Bera test for normality of the residuals was also conducted on both models. The test clearly rejects the null hypothesis of normality, a result already addressed and mitigated by introducing the t -distributed specification. In terms of accuracy, the best (i.e., lowest) values are obtained with the GARCH model, while EWMA performs worst in terms of statistical fit to the data. This indicates that GARCH provides the most consistent estimate of volatility relative to the behavior of returns. Considering the log-likelihood, the T-GARCH represents a further improvement over the standard specification, offering a more realistic treatment of extreme events. Overall, the analysis shows that among the models considered, T-GARCH offers the most balanced and statistically robust representation of volatility: it preserves the persistence captured by GARCH while improving the modeling of tail behavior. The EWMA, though less precise, remains a useful benchmark for practical applications that require a smoother, more stable measure of volatility, whereas the ARCH highlights the limitations of models with very short memory.

Question 2

The goal of this analysis is to assess the presence of time-varying volatility and correlations among the three series of daily prices. To do this, the starting point is using the returns, which are the logarithmic differences of the prices, and performing all the diagnostic statistics. What is observed is that the return plots clearly display the typical financial phenomenon of volatility clustering, as already noted earlier (Figure 13, Figure 14, Figure 15). The Ljung–Box autocorrelation test shows evidence of autocorrelation in both returns and their volatilities for the first and third series, confirmed by their correlograms, especially at the first lag (Figure 16, Figure 17,

Figure 18). Next, the Jarque–Bera normality test indicates that all residuals are non-Gaussian, a result also confirmed by the QQ plots (Figure 19, Figure 20, Figure 21). This highlights the possibility of using GARCH or DCC models with distributions different from the Gaussian. The analysis then proceeds with the EWMA model, built in the same way as in the previous case, and two graphs are presented. The first (Figure 22), showing conditional volatility, highlights that the second asset consistently exhibits higher volatility than the others. This is consistent with the high kurtosis observed during the diagnostic phase ($\approx 11, 80$), as the series displays more frequent peaks and, consequently, a higher average conditional volatility. It is also clear that all three assets experienced a sudden spike in 2020. Throughout the entire period, the three series appear to move jointly, a common feature in financial data. However, it is a model to be used with caution, as it does not have the ability to distinguish between positive and negative shocks and relies on a fixed parameter ($\lambda = 0.94$). The second graph (Figure 23) shows the correlation among the series over time. During periods of acute market stress, the correlation spikes to around 0.8–0.9, confirming the previous observation. This figure also anticipates an important result later shown with the BEKK model: the presence of spillover effects among the series, here captured through their time-varying correlations. Given the limitations of the EWMA model, the analysis continues with GARCH models. However, in their standard (1,1) specification, these models capture only part of the persistence in conditional variance, as shown by the Ljung–Box tests and the residual correlograms. This problem could be mitigated by increasing the number of lags or by adopting more flexible distribution assumptions. This is consistent with the Jarque–Bera tests on the residuals, which reject the null hypothesis of normality; therefore, the use of Student’s t-distributions is justified. In two of the three cases (for first and third assets’ returns), the estimated parameters indicate high persistence ($\alpha_1 + \beta_1 = 0,733$, $\alpha_3 + \beta_3 = 0,898$), reflecting a strong degree of conditional volatility persistence. The conditional volatility plots (Figure 24) reveal clear volatility clusters, with calm periods followed by sudden spikes typical of financial markets. Note that r_2 shows higher and more volatile conditional variance than r_1 and r_3 , indicating greater sensitivity to market shocks. All series display a sharp increase in 2020, consistent with the COVID-19 crisis, confirming that the GARCH models effectively capture market stress episodes. Ultimately, the GARCH(1,1) models capture much of the conditional dynamics but still leave room for improvement—particularly for r_1 and r_3 , where slight autocorrelation and residual ARCH effects remain. These results indicate that it is convenient to move towards multivariate models, which can better describe the joint structure of volatility. The Dynamic Conditional Correlation (DCC) model allows for the joint estimation of the univariate conditional volatilities (from the GARCH phase) and the time-varying correlations between the assets. The goal is to examine whether the interdependencies among the three return series evolve over time, responding to market shocks and revealing possible patterns of systemic co-movement. The average correlation matrix reports positive and statistically significant correlations for all pairs of returns, indicating that the assets tend to move in the same direction, with a stronger degree of co-movement between the first and the third series (Figure 25). This suggests the presence of a common systemic risk component, consistent with the patterns observed during stress periods in the EWMA and BEKK graphs. Both dynamic parameters of the DCC model are statistically significant: the first measures the reactivity of correlations to recent shocks, the latter represents their persistence over time. The sum indicates strong memory in the conditional dependence structure: the effects of shocks on correlations fade slowly, reflecting a highly persistent but non-explosive process. The final model considered in this analysis is the BEKK model. This choice is motivated by the need to obtain a more accurate representation of the dynamic structure of conditional variances and covariances among the three return series. The plots of the standardized residuals (Figure 26, Figure 27, Figure 28) show series centered around zero, with no systematic patterns and nearly constant variance over time. This indicates that the model has effectively removed the linear serial dependence present in the original returns. The ACF and PACF plots (Figure 29, Figure 30, Figure 31) visually confirm the absence of residual autocorrelation, while, as in previous cases, the residuals continue to deviate from normality. The conditional correlation matrix (Figure 32) and the spillover tests (Figure 33) show that, even in the absence of individual autocorrelation, residuals across the different series remain significantly correlated. The Ljung–Box tests on the products of residuals yield very high values ($p - value = 0.0000$), indicating persistent dynamic interdependencies among the series and a non-negligible transmission of shocks across markets. The Σ_t matrix of the BEKK model (Figure 34) represents the time evolution of conditional variances and covariances among the three returns. The plots show that the three variances (V_1, V_2, V_3) follow similar patterns, with synchronized peaks during stress periods, confirming the presence of a common volatility component. The covariances ($Cov_{12}, Cov_{13}, Cov_{23}$) are positive and highly co-moving, indicating that shocks in one market are rapidly transmitted to the others, evidence of directional spillovers and financial contagion. The correlations ($Corr_{ij}$) exhibit persistence and rise sharply during turbulent periods, consistent with the principle that correlation rises during crises. Definitely, the structure of Σ_t reflects shared systemic volatility, dominated by common risk factors rather than idiosyncratic shocks, confirming that the BEKK model effectively captures both the intertemporal and cross-market transmission of risk. Among the models considered, the BEKK model proves to be the most suitable for representing the joint dynamics of volatility and the transmission of shocks. However, in addition to the specifications used in this analysis, there is room for alternative assumptions that could improve the modulation of financial data. Possible extensions include adopting non-normal distributions, exploring different DCC specifications, or increasing the number of lags in the BEKK model (four in the case analyzed).

Question 3

As in the previous cases, we start from a diagnostic phase. Compared to before, it is worth noting that the Jarque-Bera test now confirms the normality of the oil and infl series, while normality is rejected for the growth series. It is also useful to look at the correlograms (Figure 38, Figure 39, Figure 40) of the three variables, which reveal a strong and significant autocorrelation, especially for inflation and oil prices. This result is clearly consistent with economic theory and is therefore unsurprising. The analysis continues with the selection of the optimal lag length for the VAR model. As shown in Figure 41, all information criteria point to the use of a single lag. The examination of individual equations provides results consistent with expected economic relationships. In the equation for oil prices (Figure 42), the coefficient of the variable’s own lag (oil_{t-1}) is highly significant ($p < 0.001$) and equal to 0.55, indicating huge temporal persistence in oil prices. Conversely, the lags of inflation and growth are not statistically relevant. In the inflation equation (Figure 43), the lag of inflation itself and the lag of oil are highly significant. This suggests that fluctuations in oil prices are rapidly transmitted to consumer prices, and that the inflation process exhibits substantial inertia. The growth rate, has no significant effect, confirming that short-term GDP growth does not directly drive price dynamics. In the GDP growth equation (Figure 44), the persistence of the variable is confirmed by the coefficient of $growth_{t-1}$ (0.48, $p < 0.001$). Diagnostic checks on the residuals (Figure 45, Figure 46) confirm the adequacy of the specification: the univariate Ljung–Box test for each equation finds no evidence of serial correlation, while the multivariate autocorrelation test up to the third order reports borderline significant results ($p \approx 0.02–0.04$),

but of small magnitude, suggesting only mild traces of residual dependence. The Cholesky decomposition allows the reduced-form VAR model to be transformed into a structural form, making the shocks economically interpretable. The chosen ordering of variables (*oil*, *infl*, *growth*) reflects the assumption that oil price shocks are exogenous in the short run with respect to the internal dynamics of the economy, while inflation and output growth respond with progressive lags. The lower triangular matrix shown in Figure 49 confirms this structure: the significant coefficients in the first column indicate that an innovation in the oil price not only strongly affects itself but also transmits an immediate positive effect on inflation. The impulse response functions illustrate the propagation dynamics of these shocks. A positive shock to the oil price generates an autoregressive response that fades rapidly, with the function decaying exponentially within roughly five periods (Figure 50). This points to a strong yet temporary persistence in oil prices. The effect on inflation is immediately positive, dissipating gradually within about ten periods, showing that higher energy costs are quickly passed through to consumer prices but do not produce long-term effects (Figure 51). The impact on growth (Figure 52) is negative and statistically significant in the short run: GDP experiences a contraction lasting up to the fifth period, after which the effect weakens and returns to zero, consistent with the economic notion that an oil shock acts as a temporary brake on economic activity. The reverse responses are weaker. An inflationary shock leads to a slight rise in oil prices (Figure 53) and a modest decline in growth (Figure 55), both short-lived, suggesting that inflation does not generate persistent effects on other variables. Finally, a positive shock to growth (*growth*→*oil* and *growth*→*infl*) yields weak and statistically insignificant responses, as expected given that GDP growth is not a primary driver of either oil prices or inflation. Finally, a positive unit shock to growth yields weak and statistically insignificant responses, as expected given that GDP growth is not a primary driver of either oil prices or inflation (Figure 56, Figure 57). To conclude the analysis, it can be said that the IRFs derived from the Cholesky decomposition depict an asymmetric transmission mechanism: oil price shocks are exogenous to the domestic economy and follow a clear causal sequence in which an increase in oil prices leads to higher inflation and, subsequently, to slower economic growth. The structural AB model formulation makes it possible to overcome the implicit restrictions of the Cholesky decomposition. Technically, the AB specification replaces Cholesky's purely mechanical restrictions with economically grounded constraints, built on theoretical assumptions or causal hypotheses one intends to test. For instance, the AB model allows inflation and output growth to react simultaneously (or nearly so) to an oil shock, without imposing a rigid hierarchical ordering between them. The A matrix (Figure 59) describes the system's contemporaneous structure, showing that the only statistically significant relationship is the one between inflation and oil prices, while the remaining coefficients are not significant. This means that, within the same period, an increase in oil prices immediately translates into higher price levels, whereas the effect on growth is negligible, consistent with the idea that real activity responds with delay. The B matrix (Figure 60) confirms the presence of highly significant structural variances along the diagonal, in line with the results obtained under the Cholesky identification. Note also that the Σ_t remains unchanged from the previous model (Figure 70). From an economic standpoint, the AB identification produces results largely consistent with those of the Cholesky case, though with some interesting nuances in the temporal dynamics of the impulse response functions (the response of inflation to an oil price unit shock (Figure 62) remains positive, fast, and of similar magnitude, but in the AB model the peak effect occurs slightly earlier and fades more gradually, suggesting greater inertia in the transmission of oil shocks to inflation). The reverse responses (Figure 64, Figure 67) remain weak and statistically insignificant, confirming that causality runs primarily from oil prices toward growth and inflation rather than vice versa. Compared with the Cholesky decomposition, the AB specification offers a more structural representation, where the relationship between oil prices and inflation is explicitly modeled as contemporaneous and significant, reinforcing the causal interpretation of short-run effects.

Appendix

Question 1

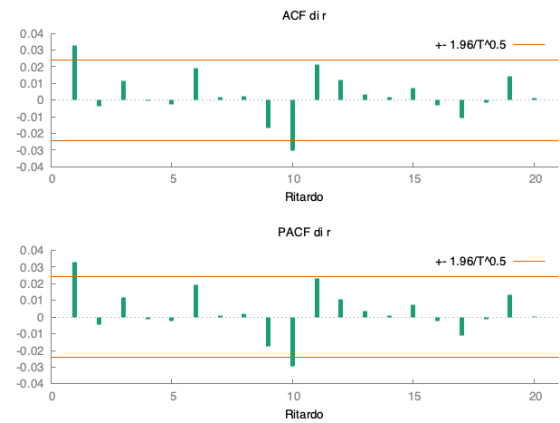


Figure 1: ACF of returns

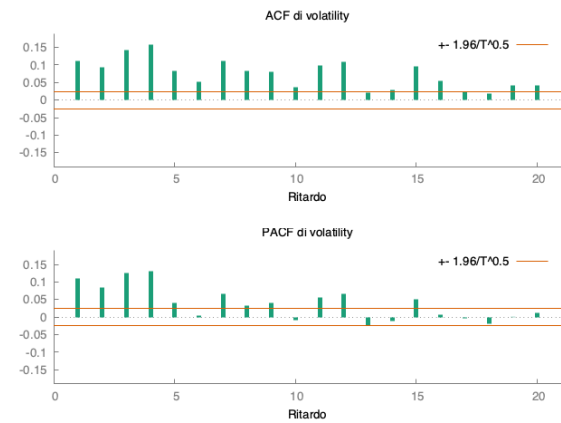


Figure 2: ACF of volatility

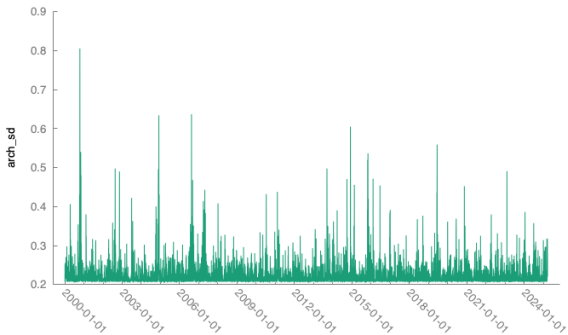


Figure 3: Conditional volatility with ARCH

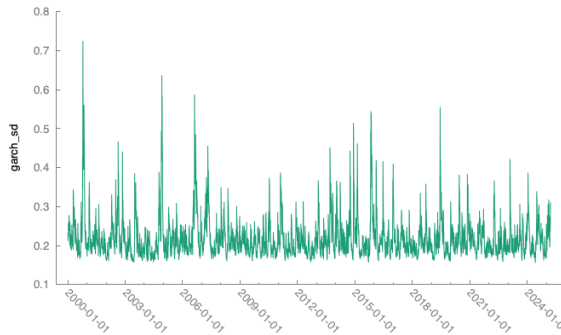


Figure 4: Conditional volatility with GARCH

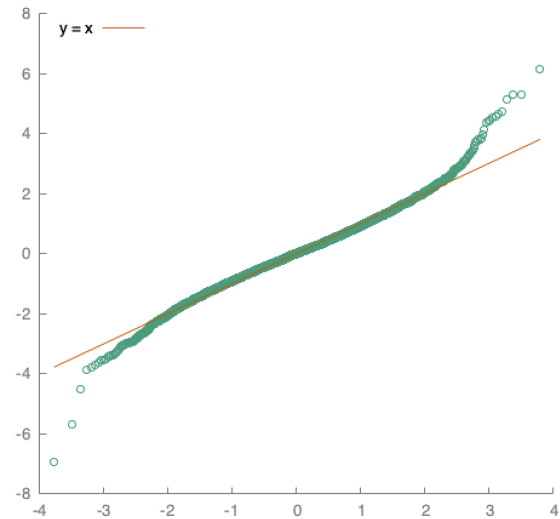


Figure 5: GARCH residuals against a Normal

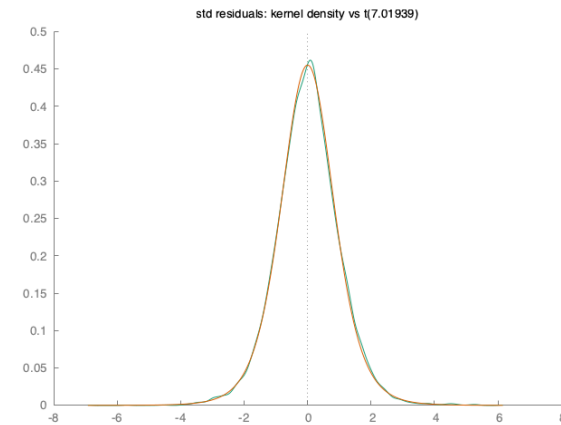


Figure 6: GARCH residuals against t-student

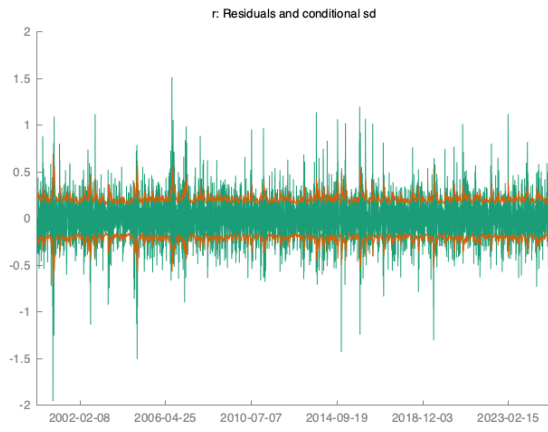


Figure 7: Residuals and volatility bands T-GARCH

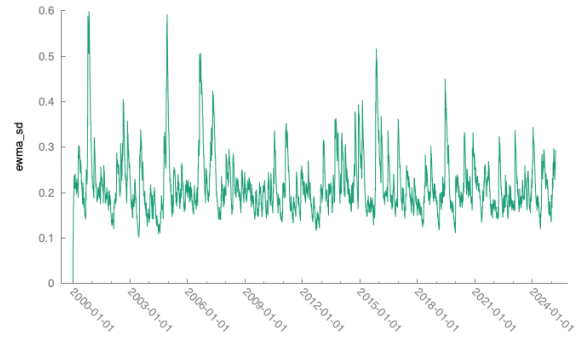


Figure 8: Conditional volatility with EWMA

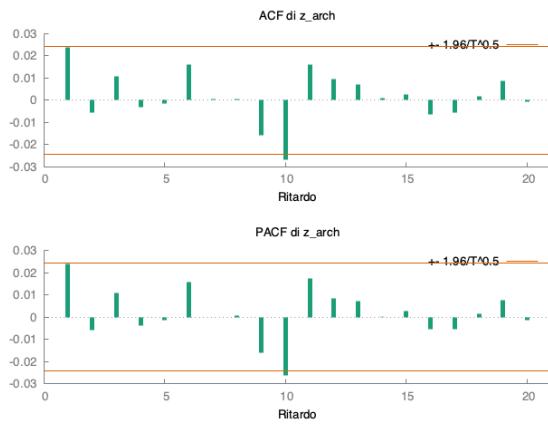


Figure 9: ACF of ARCH standardized residuals

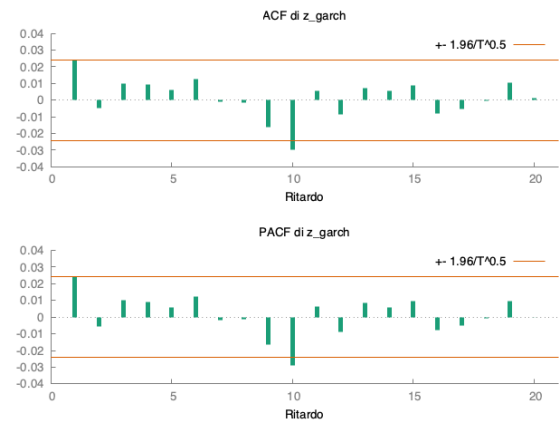


Figure 10: ACF of GARCH standardized residuals

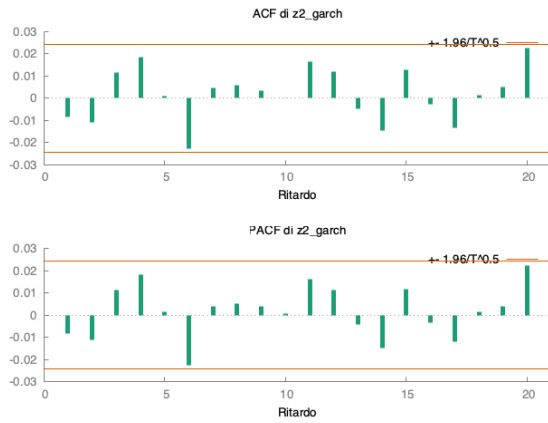


Figure 11: GARCH variance of standardized residuals

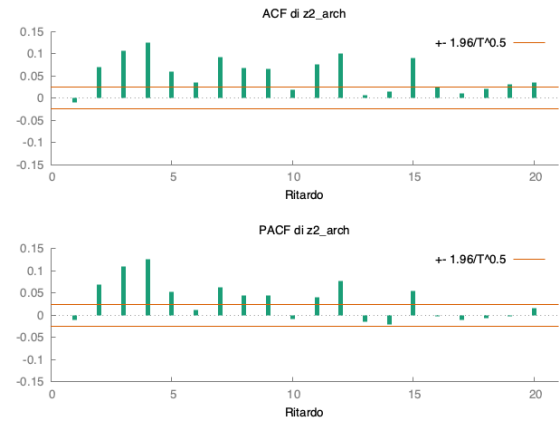


Figure 12: ARCH variance of standardized residuals

Question 2

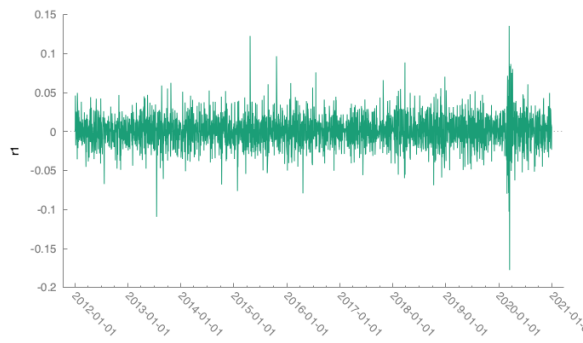


Figure 13: Returns asset 1

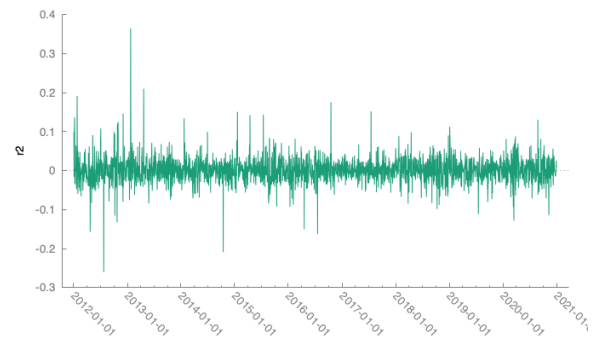


Figure 14: Returns asset 2

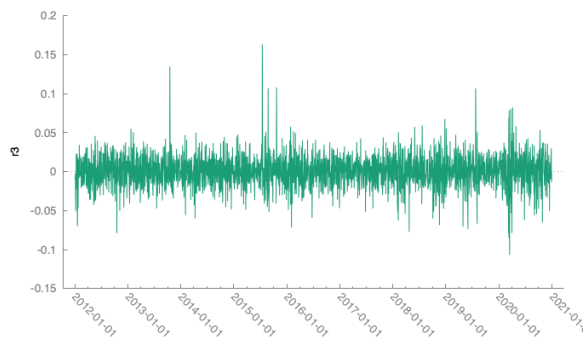


Figure 15: Returns asset 3

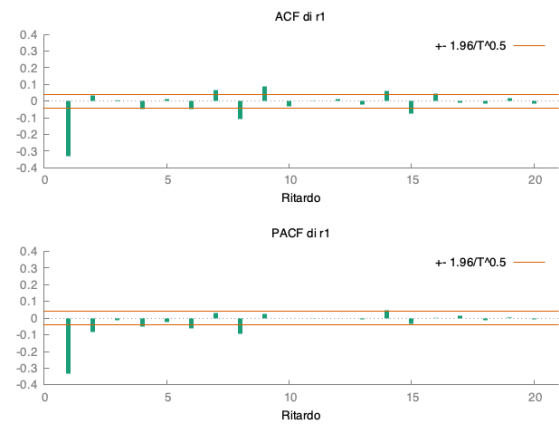


Figure 16: ACF of returns asset 1

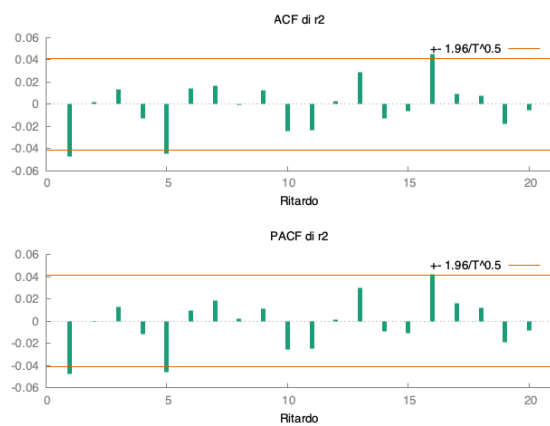


Figure 17: ACF of returns asset 2

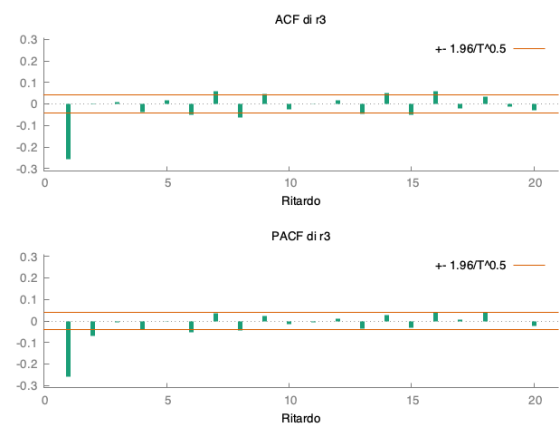


Figure 18: ACF of returns asset 3

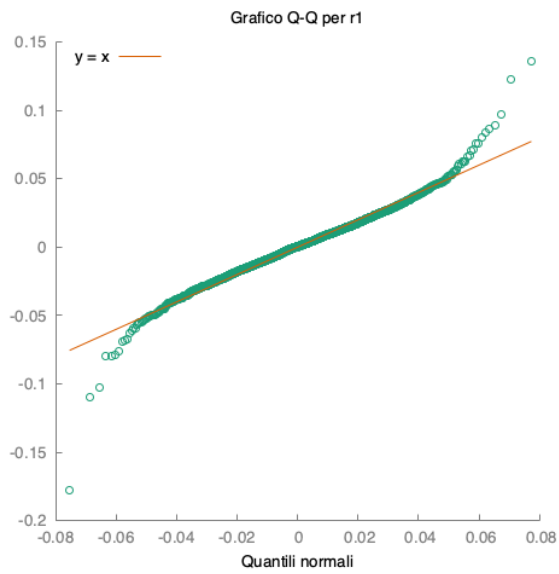


Figure 19: QQ plot asset 1

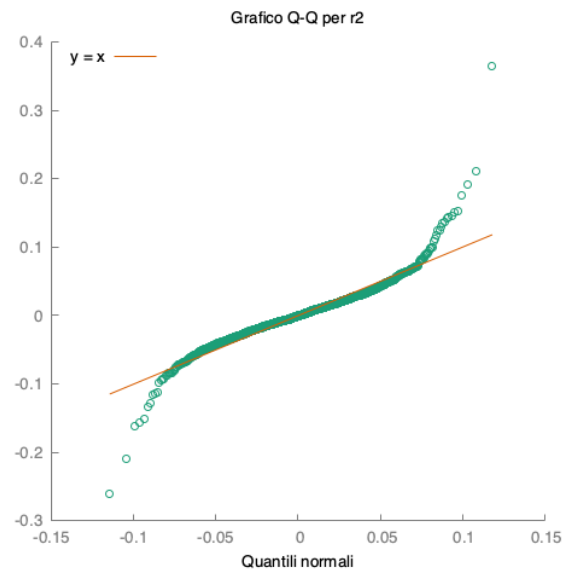


Figure 20: QQ plot asset 2

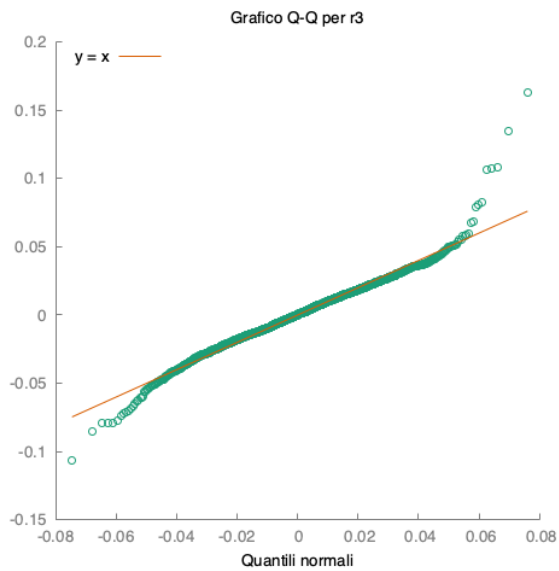


Figure 21: QQ plot asset 3

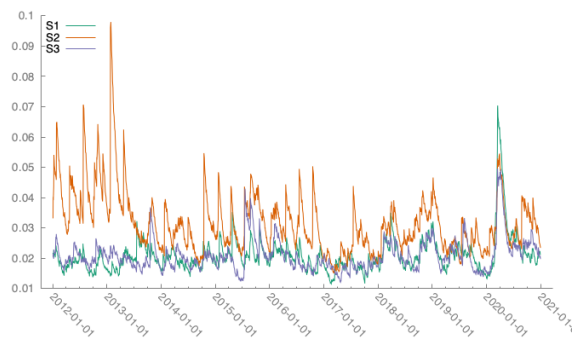


Figure 22: Standard conditional deviation EWMA

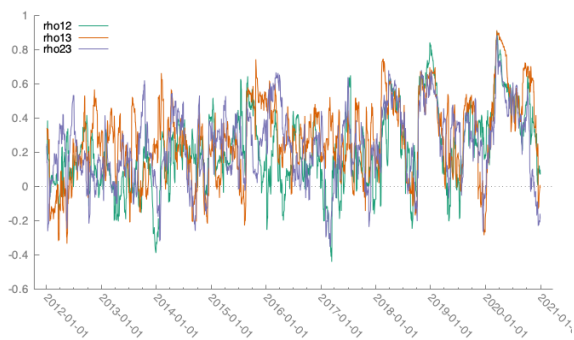


Figure 23: Correlation between assets EWMA

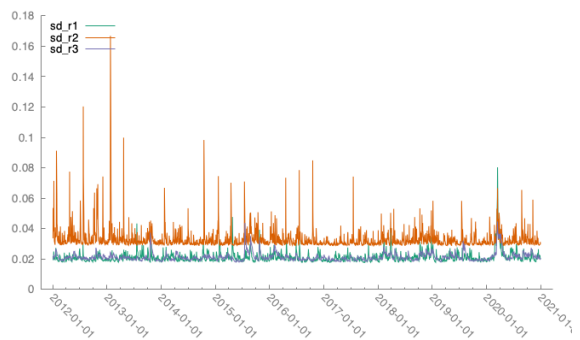


Figure 24: Conditional volatility of GARCH

Target correlation matrix:
R (3 x 3)

	et_r1	et_r2	et_r3
et_r1	1.0000	0.23859	0.34410
et_r2	0.23859	1.0000	0.25067
et_r3	0.34410	0.25067	1.0000

Scalar DCC

	coefficiente	errore std.	z	p-value	
a	0.0632596	0.0241095	2.624	0.0087	***
b	0.779422	0.116798	6.673	2.50e-11	***

Figure 25: Correlation matrix DCC

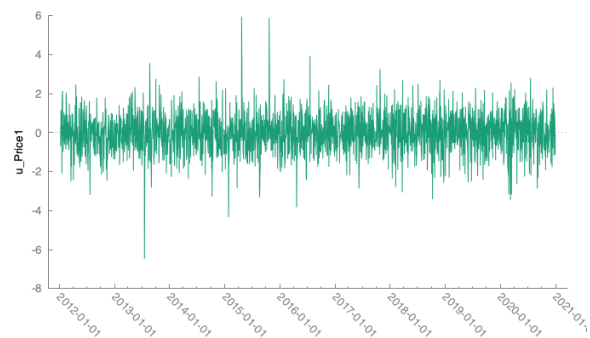


Figure 26: BEKK residuals asset 1

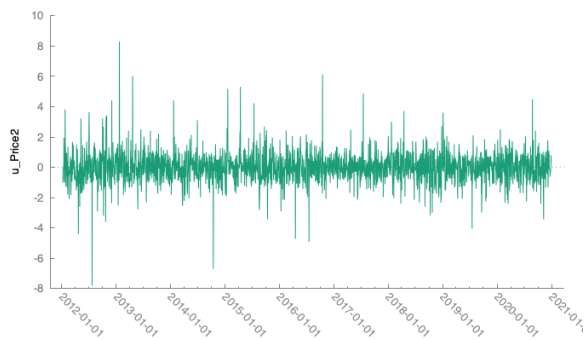


Figure 27: BEKK residuals asset 2

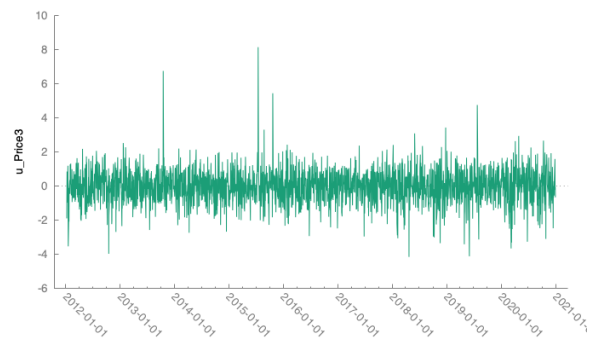


Figure 28: BEKK residuals asset 3



Figure 29: ACF BEKK residuals asset 1



Figure 30: ACF BEKK residuals asset 2

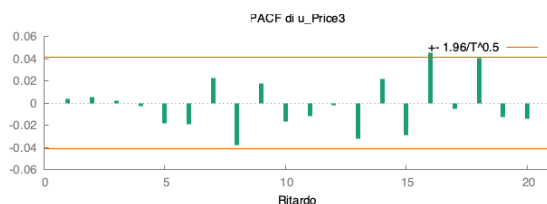
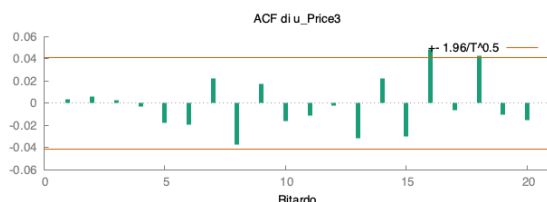
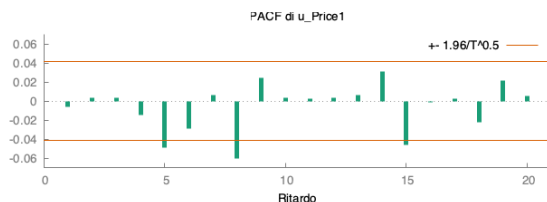


Figure 31: ACF BEKK residuals asset 3

Coefficienti di correlazione, usando le 2260 osservazioni 2012-01-10 - 2020-12-31
Valore critico a due code per n = 2260: 5% 0.0412, 1% 0.0542

	u_Price1	u_Price2	u_Price3
u_Price1	1.0000	0.2333	0.3457
u_Price2	0.2333	1.0000	0.2533
u_Price3	0.3457	0.2533	1.0000

Figure 32: BEKK residuals correlations

```

== Ljung-Box su prodotti dei residui (spillover) ==
? scalar LjungBox12 = ljungbox(u_R1_R2, 10)
? scalar Pvalue12 = pvalue(X, 10 , LjungBox12)
Ljung-Box12(10): Q = 75.8904, p-value = 0.0000
? scalar LjungBox13 = ljungbox(u_R1_R3, 10)
? scalar Pvalue13 = pvalue(X, 10 , LjungBox13)
Ljung-Box13(10): Q = 45.8633, p-value = 0.0000
? scalar LjungBox23 = ljungbox(u_R2_R3, 10)
? scalar Pvalue23 = pvalue(X, 10 , LjungBox23)
Ljung-Box23(10): Q = 48.6056, p-value = 0.0000

```

Figure 33: BEKK spillover residuals

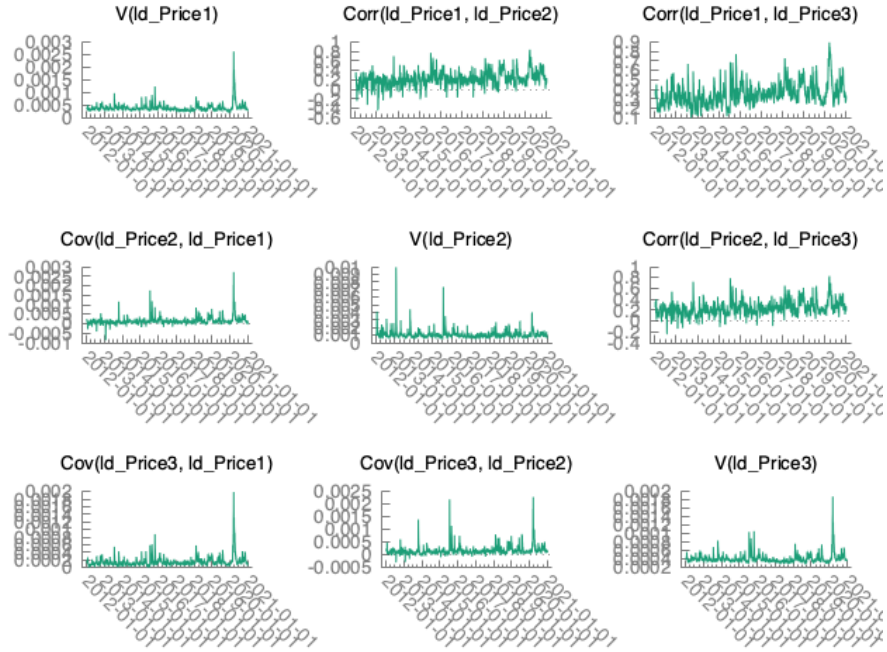


Figure 34: BEKK Σ_t

Question 3

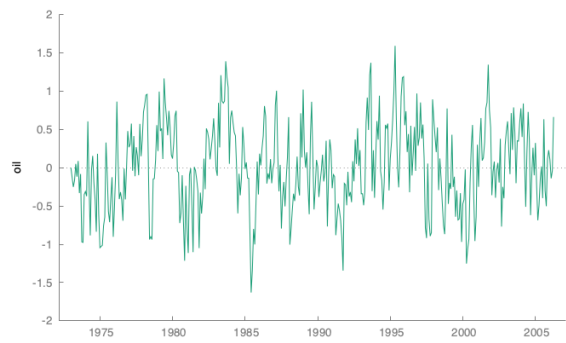


Figure 35: oil graph

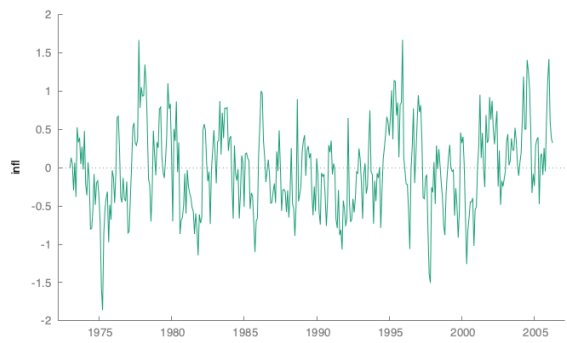


Figure 36: infl graph

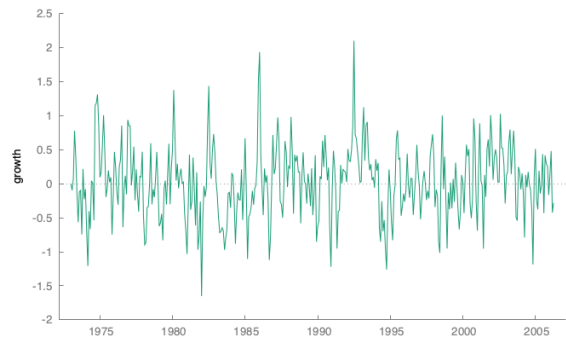


Figure 37: growth graph

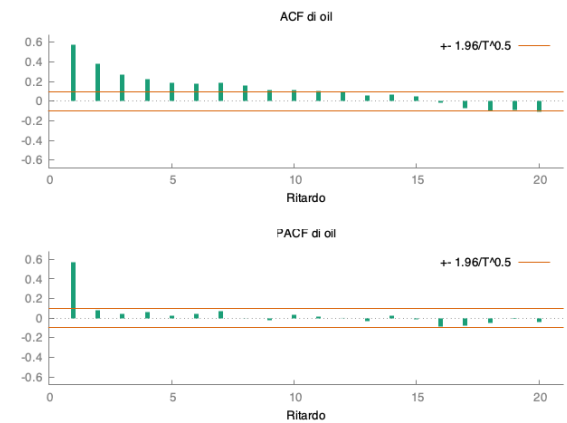


Figure 38: ACF oil

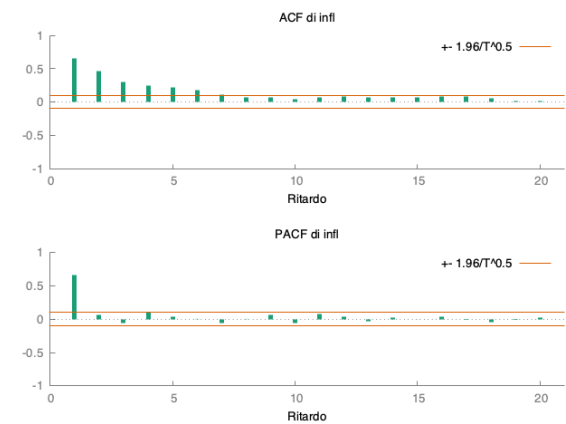


Figure 39: ACF infl

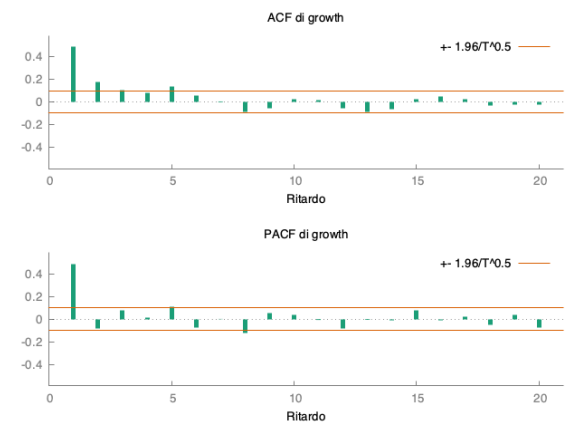


Figure 40: ACF growth

```

? modtest 3 --autocorr --univariate
Test per l'autocorrelazione fino all'ordine 3

Equazione 1:
Ljung-Box Q' = 0.919776 con p-value = P(Chi-quadro(3) > 0.919776) = 0.821

Equazione 2:
Ljung-Box Q' = 6.63449 con p-value = P(Chi-quadro(3) > 6.63449) = 0.0845

Equazione 3:
Ljung-Box Q' = 5.00812 con p-value = P(Chi-quadro(3) > 5.00812) = 0.171

```

Figure 45: Ljung-Box univariate residuals tests

```

Sistema VAR, ordine ritardi 1
Stime OLS usando le osservazioni 1973:02-2006:04 (T = 399)
Log-verosimiglianza = -681.86073
Determinante della matrice di covarianza = 0.0061225114
AIC = 3.4780
BIC = 3.5980
HQIC = 3.5255
Test portmanteau: LB(48) = 400.194, df = 423 [0.7810]

Equazione 1: oil

      coefficiente  errore std.  rapporto t  p-value
-----
const      0.00641853      0.0230995      0.2779      0.7813
oil_1       0.552629      0.0475282     11.63      4.45e-27 ***
infl_1      0.0438759      0.0481645      0.9110      0.3629
growth_1    0.0305824      0.0446559      0.6848      0.4938

Media var. dipendente  0.012608  SQM var. dipendente  0.559819
Somma quadr. residui  83.74281  E.S. della regressione  0.460442
R-quadro              0.328619  R-quadro corretto  0.323520
F(3, 395)             64.44662  P-value(F)        6.14e-34
rho                   -0.040296  Durbin-Watson     2.074508
Note: SQM = scarto quadratico medio; E.S. = errore standard

Test F per zero vincoli:

Tutti i ritardi di oil      F(1, 395) = 135.20 [0.0000]
Tutti i ritardi di infl     F(1, 395) = 0.82985 [0.3629]
Tutti i ritardi di growth   F(1, 395) = 0.46901 [0.4938]
Tutte le variabili, ritardo 1  F(3, 395) = 64.447 [0.0000]

```

Figure 42: VAR oil equation

```

Equazione 2: infl

      coefficiente  errore std.  rapporto t  p-value
-----
const      -0.00775858      0.0201847     -0.3844      0.7009
oil_1       0.234366      0.0415310      5.643      3.19e-08 ***
infl_1      0.541325      0.0420870     12.86      7.25e-32 ***
growth_1    -0.0278373      0.0390212     -0.7134      0.4760

Media var. dipendente -0.013839  SQM var. dipendente  0.555704
Somma quadr. residui  63.94246  E.S. della regressione  0.402343
R-quadro             0.479742  R-quadro corretto  0.475791
F(3, 395)            121.4129  P-value(F)        9.91e-56
rho                  -0.050336  Durbin-Watson     2.100114
Note: SQM = scarto quadratico medio; E.S. = errore standard

Test F per zero vincoli:

Tutti i ritardi di oil      F(1, 395) = 31.845 [0.0000]
Tutti i ritardi di infl     F(1, 395) = 165.43 [0.0000]
Tutti i ritardi di growth   F(1, 395) = 0.50893 [0.4760]
Tutte le variabili, ritardo 1  F(3, 395) = 121.41 [0.0000]

```

Figure 43: VAR infl equation

```

Equazione 3: growth

      coefficiente  errore std.  rapporto t  p-value
-----
const      0.0108533      0.0228138      0.4757      0.6345
oil_1       0.0418555      0.0469403      0.8917      0.3731
infl_1      -0.0933602      0.0475687     -1.963      0.0504 *
growth_1    0.478164      0.0441036     10.84      3.76e-24 ***

Media var. dipendente  0.024942  SQM var. dipendente  0.521731
Somma quadr. residui  81.68399  E.S. della regressione  0.454747
R-quadro             0.246019  R-quadro corretto  0.240292
F(3, 395)            42.96188  P-value(F)        4.77e-24
rho                  0.045294  Durbin-Watson     1.909247
Note: SQM = scarto quadratico medio; E.S. = errore standard

Test F per zero vincoli:

Tutti i ritardi di oil      F(1, 395) = 0.79509 [0.3731]
Tutti i ritardi di infl     F(1, 395) = 3.8519 [0.0504]
Tutti i ritardi di growth   F(1, 395) = 117.55 [0.0000]
Tutte le variabili, ritardo 1  F(3, 395) = 42.962 [0.0000]

```

Figure 44: VAR growth equation

```

? modtest 3 --autocorr
Test per l'autocorrelazione fino all'ordine 3

      Rao F    Approx dist.  p-value
lag 1      2.157      F(9, 949)      0.0229
lag 2      1.638      F(18, 1095)     0.0448
lag 3      1.688      F(27, 1122)     0.0156

```

Figure 46: Ljung-Box vector of residuals test

```
? modtest 3 --arch --univariate
Test per ARCH di ordine 3
```

Equazione 1:

	coefficiente	errore std.	rapporto t	p-value
alpha(0)	0.247418	0.0236287	10.47	8.67e-23 ***
alpha(1)	-0.0386764	0.0502980	-0.7689	0.4424
alpha(2)	-0.0411882	0.0503033	-0.8188	0.4134
alpha(3)	-0.0918178	0.0502942	-1.826	0.0687 *

Ipotesi nulla: non sono presenti effetti ARCH
 Statistica test: LM = 4.36968
 con p-value = $P(\text{Chi-quadro}(3) > 4.36968) = 0.224213$

Equazione 2:

	coefficiente	errore std.	rapporto t	p-value
alpha(0)	0.166265	0.0179500	9.263	1.33e-18 ***
alpha(1)	0.0536169	0.0504134	1.064	0.2882
alpha(2)	-0.0192555	0.0504716	-0.3815	0.7030
alpha(3)	-0.0654125	0.0504051	-1.298	0.1951

Ipotesi nulla: non sono presenti effetti ARCH
 Statistica test: LM = 3.04429
 con p-value = $P(\text{Chi-quadro}(3) > 3.04429) = 0.384847$

Equazione 3:

	coefficiente	errore std.	rapporto t	p-value
alpha(0)	0.203323	0.0239022	8.506	3.83e-16 ***
alpha(1)	0.0496707	0.0504368	0.9848	0.3253
alpha(2)	-0.0697931	0.0503633	-1.386	0.1666
alpha(3)	0.0271374	0.0504111	0.5383	0.5907

Ipotesi nulla: non sono presenti effetti ARCH
 Statistica test: LM = 2.94533
 con p-value = $P(\text{Chi-quadro}(3) > 2.94533) = 0.400132$

Figure 47: ARCH univariate tests

	coefficiente	errore std.	z	p-value
C[1; 1]	0.459861	0.0162585	28.28	5.40e-176 ***
C[2; 1]	0.130150	0.0195577	6.655	2.84e-11 ***
C[3; 1]	-0.0326668	0.0226792	-1.440	0.1498
C[1; 2]	0.00000	0.00000	NA	NA
C[2; 2]	0.380173	0.0134412	28.28	5.40e-176 ***
C[3; 2]	-0.0174099	0.0226414	-0.7689	0.4419
C[1; 3]	0.00000	0.00000	NA	NA
C[2; 3]	0.00000	0.00000	NA	NA
C[3; 3]	0.452662	0.0160040	28.28	5.40e-176 ***

Figure 49: Cholesky lower triangular matrix

```
? modtest 3 --arch
Test per ARCH di ordine 3
```

	LM	df	p-value
lag 1	32.903	36	0.6167
lag 2	68.482	72	0.5957
lag 3	99.043	108	0.7195

Figure 48: ARCH test on vector of residuals

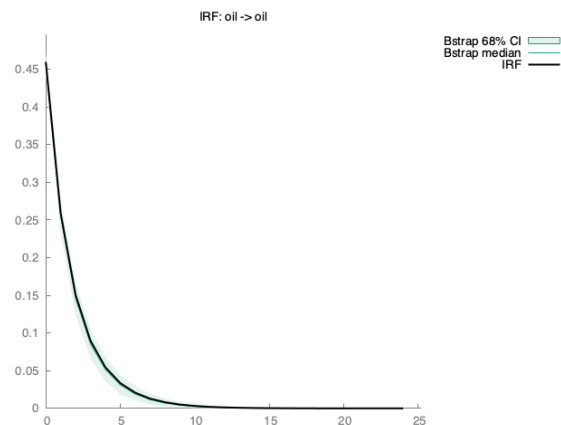


Figure 50: IRF oil-oil with Cholesky

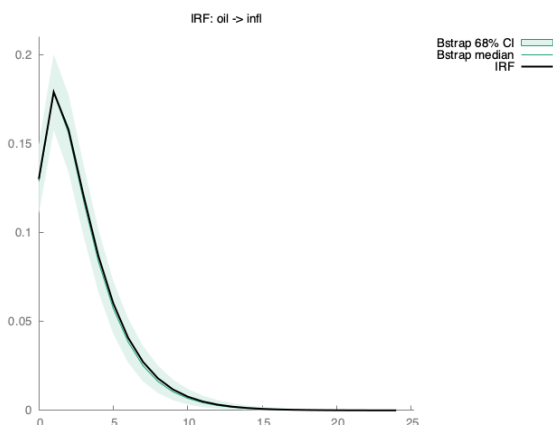


Figure 51: IRF oil-infl with Cholesky

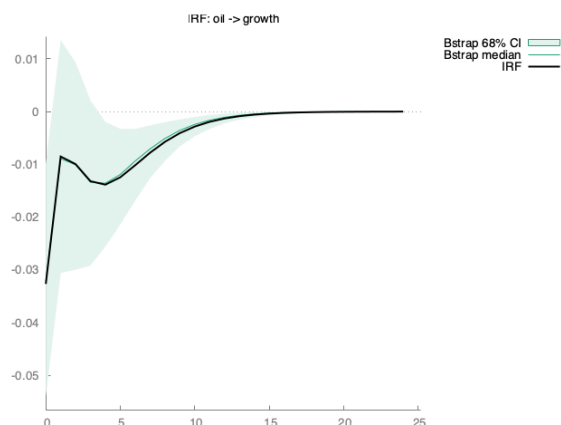


Figure 52: IRF oil-growth with Cholesky

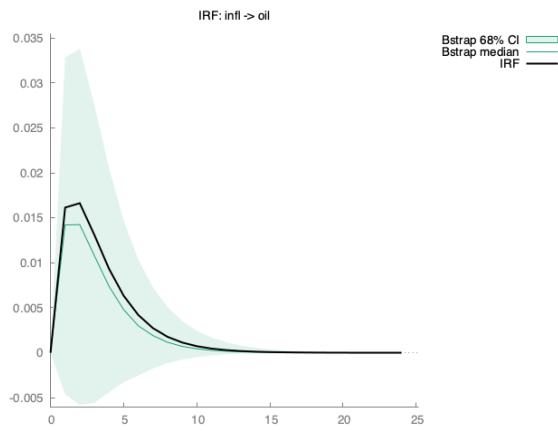


Figure 53: IRF infl-oil with Cholesky

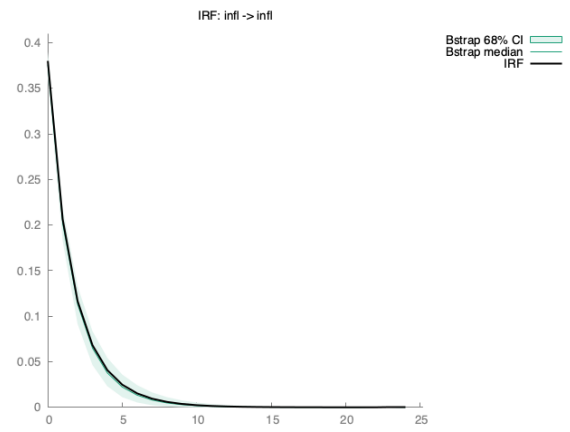


Figure 54: IRF infl-infl with Cholesky

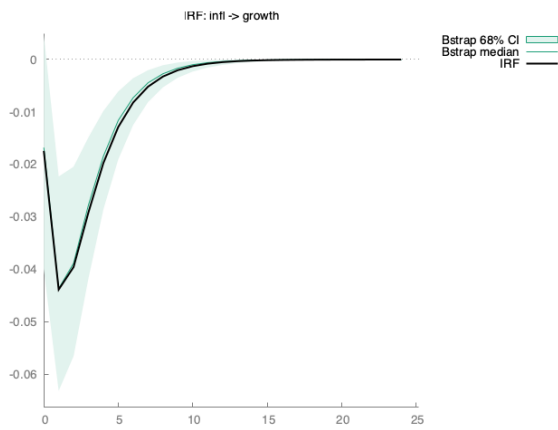


Figure 55: IRF infl-growth with Cholesky

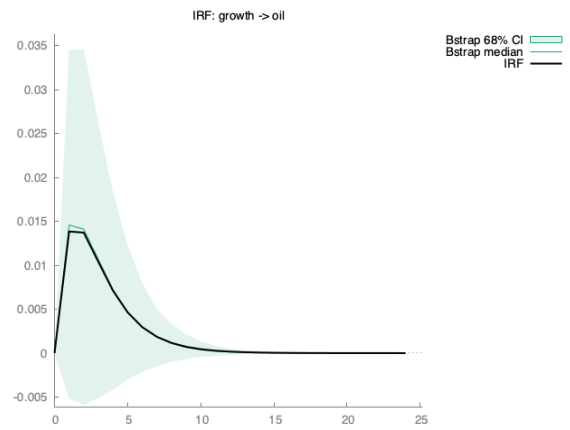


Figure 56: IRF growth-oil with Cholesky

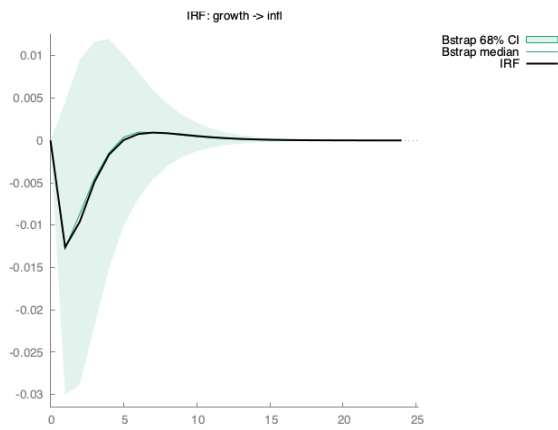


Figure 57: IRF growth-infl with Cholesky

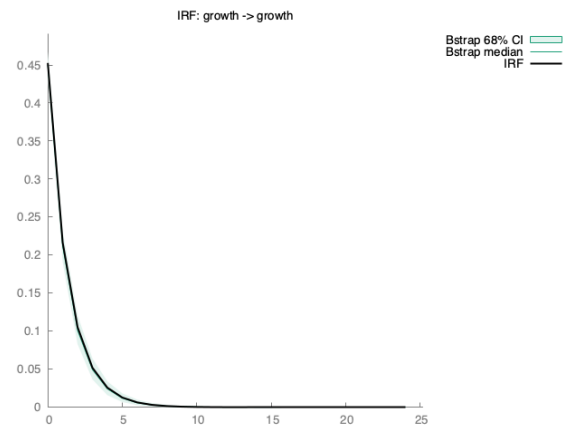


Figure 58: IRF growth-growth with Cholesky

	coefficiente	errore std.	z	p-value
A[1; 1]	1.00000	0.00000	NA	NA
A[2; 1]	-0.280730	0.0414124	-6.779	1.21e-11 ***
A[3; 1]	0.0710364	0.0492536	1.442	0.1492
A[1; 2]	0.00000	0.00000	NA	NA
A[2; 2]	1.00000	0.00000	NA	NA
A[3; 2]	0.00000	0.00000	NA	NA
A[1; 3]	0.00000	0.00000	NA	NA
A[2; 3]	0.0322544	0.0419311	0.7692	0.4418
A[3; 3]	1.00000	0.00000	NA	NA

Figure 59: A Matrix AB decomposition

	coefficiente	errore std.	z	p-value
B[1; 1]	0.459861	0.0162585	28.28	5.40e-176 ***
B[2; 1]	0.00000	0.00000	NA	NA
B[3; 1]	0.00000	0.00000	NA	NA
B[1; 2]	0.00000	0.00000	NA	NA
B[2; 2]	0.379892	0.0134312	28.28	5.40e-176 ***
B[3; 2]	0.00000	0.00000	NA	NA
B[1; 3]	0.00000	0.00000	NA	NA
B[2; 3]	0.00000	0.00000	NA	NA
B[3; 3]	0.452996	0.0160158	28.28	5.40e-176 ***

Log-likelihood = -688.098

Figure 60: B Matrix AB decomposition

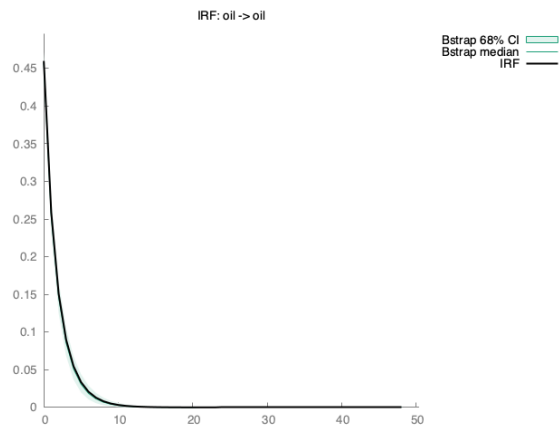


Figure 61: IRF oil-oil with AB

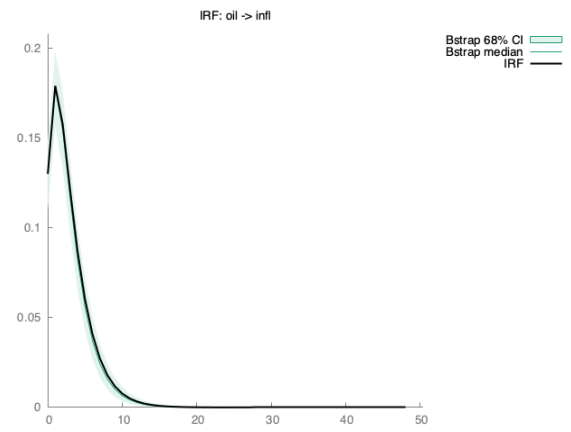


Figure 62: IRF oil-infl with AB

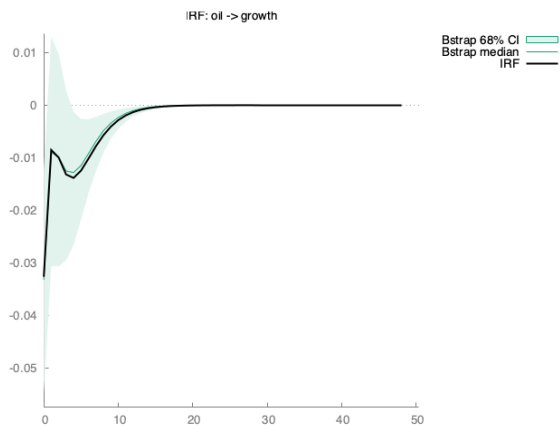


Figure 63: IRF oil-growth with AB

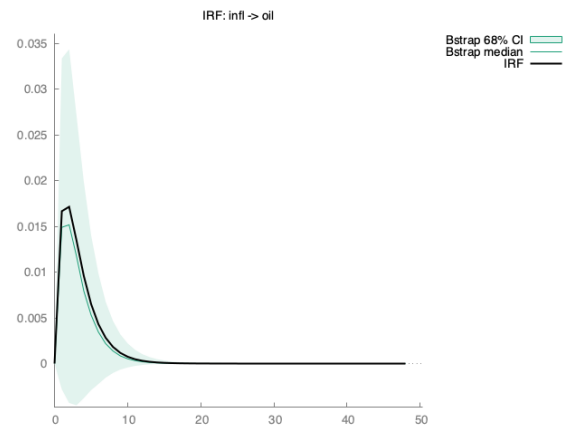


Figure 64: IRF infl-oil with AB

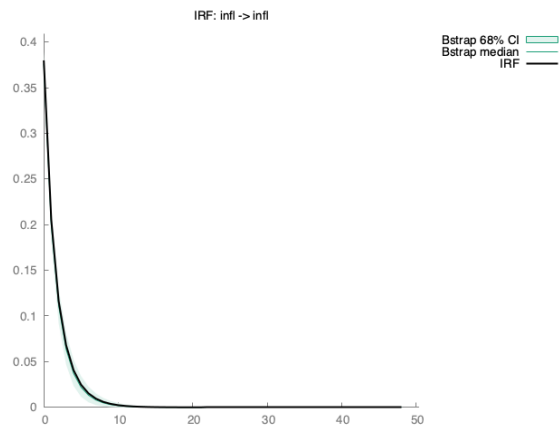


Figure 65: IRF infl-infl with AB

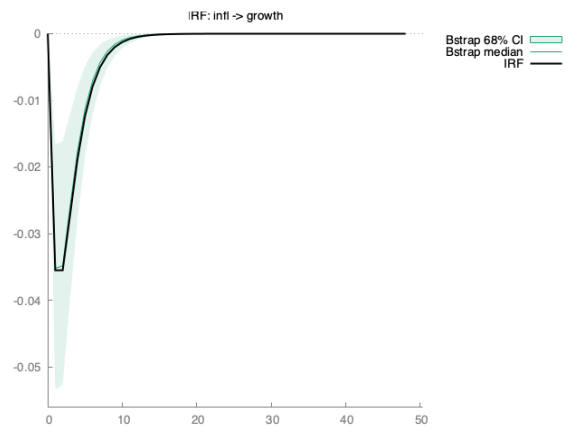


Figure 66: IRF infl-growth with AB

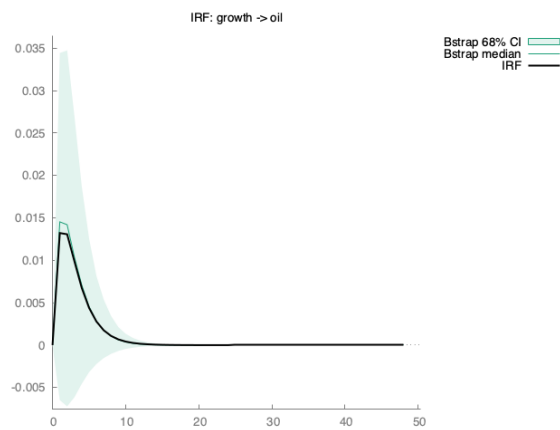


Figure 67: IRF growth-oil with AB

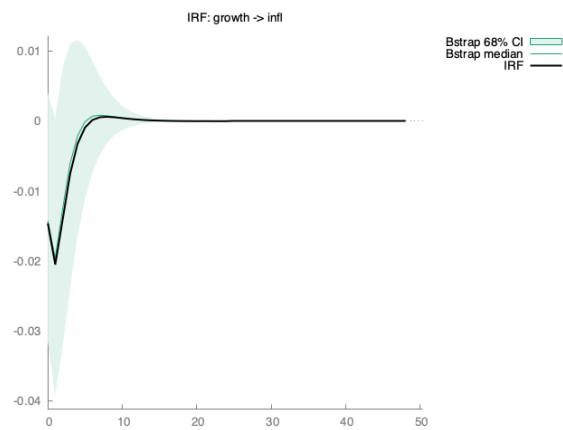


Figure 68: IRF growth-infl with AB

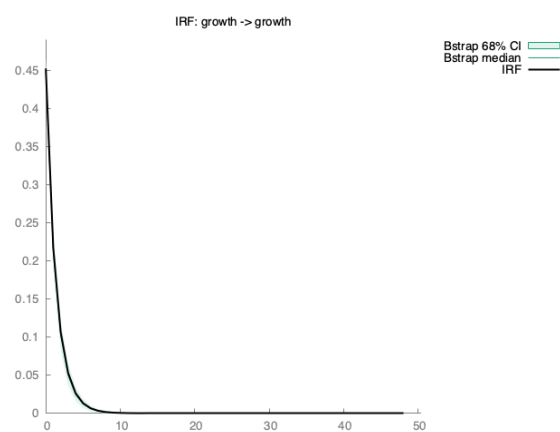


Figure 69: IRF growth-growth with AB

Unconstrained Sigma:

0.21147	0.05985	-0.01502
0.05985	0.16147	-0.01087
-0.01502	-0.01087	0.20627

Figure 70: AB unconstrained Σ_t matrix

Here's the link of the Gretl code: <https://github.com/alessandrobonfiglio/Econometrics-of-financial-markets.git>